higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

DIGITAL ELECTRONICS N4

13 July 2022

This marking guideline consists of 9 pages.

SECTION A

QUESTION 1

1.1 D
1.2 B
1.3 D
1.4 A
1.5 D

(5 × 2) **[10]**

QUESTION 2

2.1 False
2.2 False
2.3 False
2.4 False
2.5 False
2.6 True
2.7 True
2.8 False

(8 × 1) **[8]**

QUESTION 3

3.1 D
3.2 G
3.3 B
3.4 A
3.5 C
3.6 E
3.7 F

(7 × 1) **[7]**

TOTAL SECTION A: 25

QUESTION 4

4.1

BINARY

	0	*	0	0	1	0	1	1	,	0	1	1	0
	1	*	0	1	1	0	0	0	,	0	1	0	1

1'S COMPLEMENT

	0	*	0	0	1	0	1	1	,	0	1	1	0
	1	*	1	0	0	1	1	1	,	1	0	1	0
0	1	*	1	1	0	0	1	1	,	0	0	0	0
													0
	1	*	1	1	0	0	1	1	,	0	0	0	0

NEG 1 1 0 0 , 1 1 1 1 $_2$

Allocate 1 mark for each of the following steps indicated correctly.

	0	*	0	0	1	0	1	1	,	0	1	1	0
	1	*	1	0	0	1	1	1	,	1	0	1	0
0	1	*	1	1	0	0	1	1	,	0	0	0	0
	1	*	1	1	0	0	1	1	,	0	0	0	0

NEG 1 1 0 0 , 1 1 1 1 $_2$

√ √ √ √ √

4.2

BINARY

0 * 0 1 0 0 1 , 1 1 0
1 * 0 1 1 1 0 , 0 1 1

2'S COMPLEMENT

0 * 0 1 0 0 1 , 1 1 0
1 * 1 0 0 0 1 , 1 0 1
0 1 * 1 1 0 1 1 , 0 1 1

0 0 1 0 0 , 1 0 0
Plus 1 1
NEG 0 1 0 0 , 1 0 1 $_2$

Allocate 1 mark for each of the following steps indicated correctly.

0 * 0 1 0 0 1 , 1 1 0 ✓

1 * 1 0 0 0 1 , 1 0 1 ✓

0 1 * 1 1 0 1 1 , 0 1 1 ✓

NEG ✓ 0 1 0 0 , 1 0 1 $_2$ ✓

4.3

1 0 1 1 0, 0 1 $_2$
mult 1 0 0 1, 0 1 $_2$

1 0 1 1 0 0 1
1 0 1 1 0 0 1
1 0 1 1 0 0 1
1 1 0 0 1 1 0 1, 1 1 0 1 $_2$

Allocate 2 marks if it was indicated that the following number must be added.

1 0 1 1 0 0 1
1 0 1 1 0 0 1
1 0 1 1 0 0 1

✓ ✓

Allocate 2 marks if the answer is correct.

1 1 0 0 1 1 0 1, 1 1 0 1 $_2$ ✓ ✓

Allocate 1 mark if the comma is in the correct position.

1 1 0 0 1 1 0 1, 1 1 0 1 $_2$ ✓

4.4

```
                                    1  1  0  0,  1  0  1 2
          1  1,  0  1  1 | 1  0  1  0  1  0,  1  0  0  1  1  1
                           1  1  0  1  1
                           ------------------------------------
                              1  1  1  1  1
                              1  1  0  1  1
                              ---------------------------------
                                 1  0  0  0  0  1
                                 1  1  0  1  1
                                 ---------------------------
                                    1  1  0  1  1
                                    1  1  0  1  1
                                    ---------------------------
                                                0
```

Allocate 1 mark for each of the following steps indicated correctly

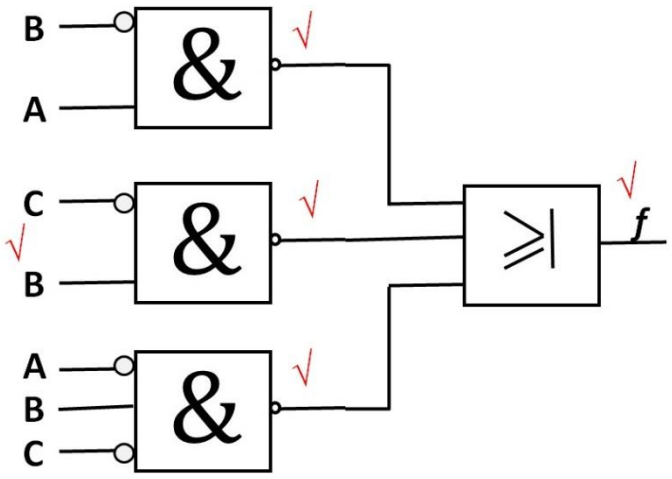
```
1     1     1     1     1
1  0  0  0  0  1                    ✓
1     1     0     1     1           ✓
1  1  0  0,  1  0  1 2              ✓
                                    ✓
```

Allocate 1 mark if the comma is in the correct position. ✓

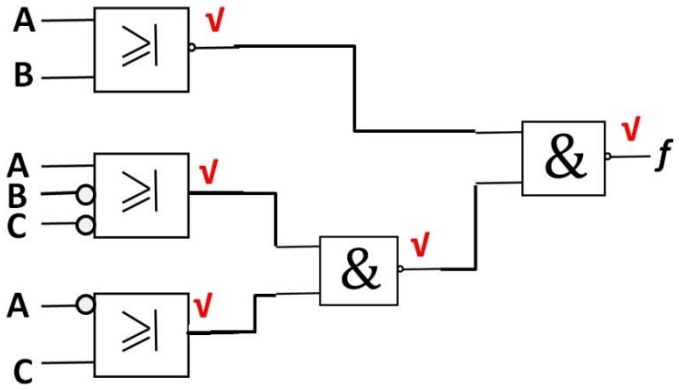
(4 × 5) **[20]**

QUESTION 5

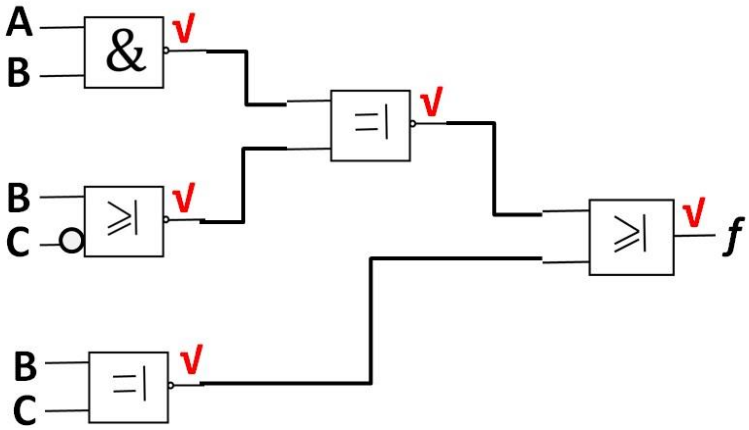
5.1 $\overline{A\overline{B}} + \overline{B\overline{C}} + \overline{\overline{A}B\overline{C}} = f$



5.2 $\overline{(\overline{A+B})\overline{(A+\overline{B}+\overline{C})(\overline{A}+C)}} = f$



5.3 $\overline{\overline{A\overline{B}} \oplus \overline{\overline{B+\overline{C}}} + \overline{B \oplus C}} = f$



(3 × 5) **[15]**

QUESTION 6

6.1 $(\overline{A}C + B)(B + \overline{C})(AB + C)(A + \overline{B}) = f$

$(\overline{A}BC + AC\overline{C} + BB + B\overline{C})(AAB + AB\overline{B} + AC + \overline{B}C) = f$

$(\overline{A}BC + 0 + B + B\overline{C})(AB + 0 + AC + \overline{B}C) = f$

$(B + B\overline{C})(AB + AC + \overline{B}C) = f$ ✓

$(B) \bullet (AB + AC + \overline{B}C) = f$ ✓

$(ABB + ABC + \overline{B}BC) = f$ ✓

$(AB + ABC + 0) = f$

$AB + ABC = f$ ✓

$AB = f$ ✓

6.2 $\overline{\overline{(\overline{A}BC + A\overline{B}C)}\,\overline{(AB\overline{C} + ABC)}} = f$

$\overline{\overline{(\overline{A}BC + A\overline{B}C)}} + \overline{\overline{(AB\overline{C} + ABC)}} = f$ ✓

$(\overline{A}BC + A\overline{B}C) + (AB\overline{C} + ABC) = f$

$\overline{A}BC + A\overline{B}C + AB\overline{C} + ABC = f$

$\overline{A}BC + A\overline{B}C + AB(\overline{C} + C) = f$

$\overline{A}BC + A\overline{B}C + AB = f$

$\overline{A}BC + A \bullet (\overline{B}C + B) = f$ ✓

$\overline{A}BC + A \bullet (C + B) = f$ ✓

$\overline{A}BC + AC + AB = f$

$(\overline{A}B + A)C + AB = f$ ✓

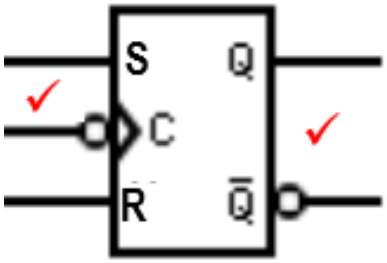
$(B + A)C + AB = f$

$BC + AC + AB = f$ ✓

(2 × 5) **[10]**

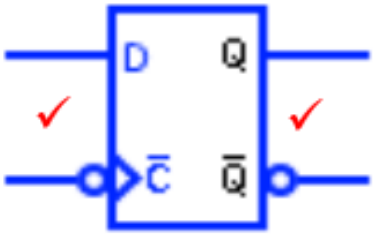
QUESTION 7

7.1 7.1.1 SR



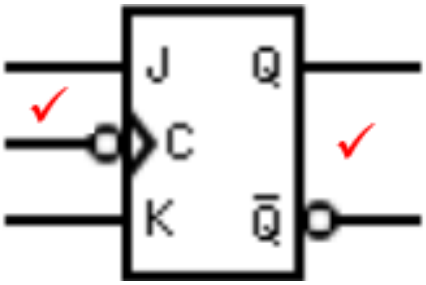
S	R	Q	$\overline{Q}$	
0	0	UNCHANGED		√
0	1	0	1	
1	0	1	0	√
1	1	UNDEFINED		

7.1.2 D-TYPE



D	Q	$\overline{Q}$	
0	0	1	√
1	1	0	√

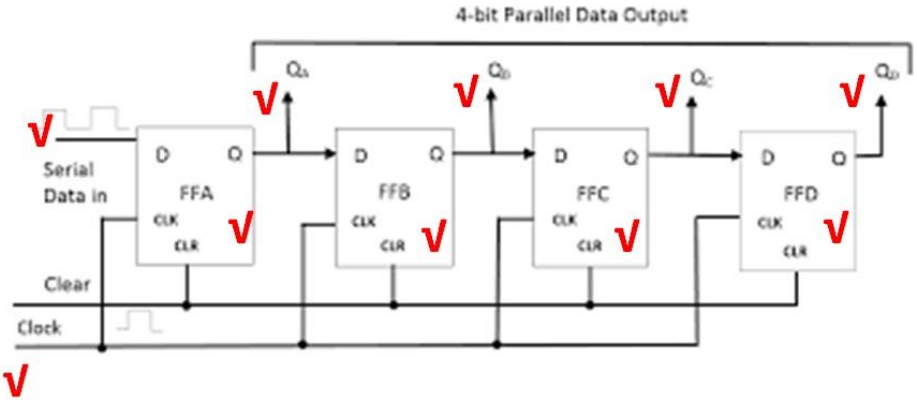
7.1.3 JK



J	K	Q	$\overline{Q}$	
0	0	UNCHANGED		√
0	1	0	1	
1	0	1	0	√
1	1	TOGGLE		

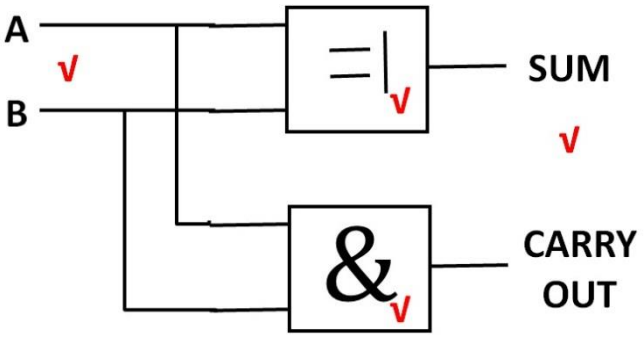
(3 × 4) (12)

7.2



(10)

7.3



A	B	SUM	C/OUT	
0	0	0	0	√
0	1	1	0	√
1	0	1	0	√
1	1	0	1	√

(8)
[30]

TOTAL SECTION B: 75
GRAND TOTAL: 100